Lesson Modifications

This lesson can be easily modified depending on student needs. More locations can be added to the graph to provide more practice. Modifying the shape of the geofenced areas can change the difficulty level of graphing them.

One version of this activity has all coordinates in quadrant 1 and another version has some coordinates in each of the four quadrants. The answers for both versions can be found in this educator guide.

Locations on Earth can all be specified using latitude and longitude. As an enrichment activity, students can research longitude and latitude using the internet or other resources. Comparing and contrasting longitude and latitude to the coordinate system in this exercise helps students further understood the real-world application of graphing while mapping. Have them determine where on Earth would be the origin in the latitude/longitude system.

Grouping Students

This activity is designed to be completed independently. Grouping can, however, be used if necessary. Students working in pairs can independently solve the problem and then compare answers to validate their solutions or find errors.

Steps

1. Show the students the NASA video about Safeguard (technology used to implement geofencing). The video is available at: *https://www.youtube.com/watch?v=0Kc01cV7vCU*.
2. Review graphing on the Cartesian coordinate system.
3. Show the map of Cheyenne, Wyoming without the grid. Discuss the function of air taxis and the dangers associated with flying in some areas, such as over the airport.
4. Show the map of Cheyenne, Wyoming with the grid. Discuss how using the grid makes it possible to specify locations because it provides a reference point.
5. Distribute the student guides and allow students to begin working.
6. Allow students to verify answers for the “Getting the Coordinates” section. (This allows students to find errors that would be compounded in the “Using the Distance Formula” section.)